

Shubham Kulkarni

AI Engineer | Computer Vision | Generative AI | MLOps | Edge AI
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PROFESSIONAL SUMMARY

Production-focused **AI Engineer** specializing in **Computer Vision**, **Generative AI**, and **MLOps**. Proven track record of deploying **scalable, end-to-end systems** that process over **150,000 events daily** with **95% accuracy**. Expertise spans **edge AI deployment** (NVIDIA Jetson), **low-latency inference** (30% reduction), and **multi-modal RAG architectures**. Recognized nationally as a **Smart India Hackathon winner (AIR 1)** for developing high-impact AI solutions.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS

AI / ML: Computer Vision, Deep Learning, NLP, Large Language Models (LLMs), Hugging Face Transformers

Frameworks: YOLOv5/v8, OpenCV, PyTorch, TensorFlow, PaddleOCR, ByteTrack, OSNet, ResNet

Generative AI: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, FAISS, Vector Databases, Embeddings

MLOps & Backend: MLOps, FastAPI, Flask, Django, REST APIs, Async Systems, Microservices, Docker, CI/CD, Git

Model Deployment: TensorRT, ONNX, Model Optimization, Inference Pipeline Tuning, Batch Processing, Multi-threading

Edge AI & Hardware: NVIDIA Jetson Nano, NVIDIA Jetson Xavier, Raspberry Pi, Rockchip

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP)

Databases & Caching: PostgreSQL, Redis, MSSQL

PROFESSIONAL EXPERIENCE

Software Engineer – AI/ML

Aug 2025 – Present

Arya Omnitalk Wireless Solutions Pvt. Ltd. | Pune, Maharashtra | Full-time, On-site

- Built and scaled two production-grade AI and Computer Vision pipelines (VIDES and ATMS), achieving 95% model accuracy and processing 50,000+ events/day in live environments.
- Developed real-time object detection and multi-object tracking models using YOLOv8, OpenCV, and ByteTrack for vehicle detection, classification, and traffic violation analysis with under 100ms inference latency.
- Engineered full end-to-end systems integrating NVIDIA Jetson Nano/Xavier edge devices with backend microservices via async REST APIs for real-time field deployment.
- Optimized inference pipelines using multi-threading and batch processing, reducing end-to-end system latency by 30% in production — directly improving real-time responsiveness for field operators.
- Built OCR-based structured data extraction pipelines using PaddleOCR, reducing manual data entry effort by eliminating manual reporting workflows for the operations team.
- Collaborated with hardware engineers, backend developers, and operations stakeholders across 3 teams to integrate AI capabilities into existing production infrastructure.
- Developed and maintained FastAPI and Flask backend services including real-time dashboards for system monitoring and operator interaction.
- Deployed and maintained AI systems across cloud (AWS/Azure) and edge environments, ensuring 99% uptime and continuous delivery via CI/CD pipelines.

PROJECTS

Vehicle Forensic Matching System

Github: https://github.com/kulkarnishub377/Vechile_forensic_match_RAC

PyTorch | OpenCV | FAISS | FastAPI | Redis | PaddleOCR | ResNet | OSNet

- Problem:** Manual vehicle identification from surveillance footage was slow and error-prone across large databases of 150,000+ entries.

- Built a multi-modal vehicle retrieval system combining deep learning embeddings (ResNet + OSNet), OCR-extracted license plate data, and vector search — enabling automated forensic identification at scale.
- Optimized retrieval speed by 30 times using FAISS IVF indexing, reducing average query response time to under 1 second and cutting investigator lookup time significantly.
- Scaled to handle 150,000+ vehicle queries/day with 99% uptime via async processing, Redis caching, and load-balanced FastAPI services.

Document AI + Retrieval-Augmented Generation (RAG) Pipeline

Github: <https://github.com/kulkarnishub377/Document-AI---RAG-Pipeline>

PyTorch | LangChain | FAISS | PaddleOCR | LLM | RAG

- Problem: Teams spent hours manually reviewing large volumes of PDFs and scanned documents to extract key information.
- Built an end-to-end Document AI pipeline using OCR and layout analysis to extract structured data from PDFs and scanned images, eliminating manual document review.
- Implemented a full RAG architecture with semantic chunking, vector indexing, and LLM-based reasoning — enabling natural language Q&A over documents with sub-2 second response time.
- Designed reusable prompt pipelines for summarization, key-field extraction, and table parsing — reducing document processing time by enabling instant querying over large document corpora.

ACHIEVEMENTS & AWARDS

- **AIR 1 — Smart India Hackathon 2023 (Team Lead)** — Developed an [AI-based e-waste management](#) system for automated waste classification. Selected as best solution from thousands of teams across India.
- **University Rank 2 — SPPU Startup Olympiad 2025** — Built an [AI smart irrigation system](#) (live [JalTantra](#)) using predictive modelling to minimize water usage and maximize crop yield.
- **Runner-Up — AI Hackathon, TIAA Global Pvt. Ltd.** — Developed an AI-powered investment intelligence platform with risk-based scoring and data-driven financial recommendations.

EDUCATION

Bachelor of Engineering (B.E.) — Electronics & Telecommunication

2021 – 2025

Dr. Vithalrao Vikhe Patil College of Engineering | Ahilyanagar (Ahmednagar), Maharashtra | Savitribai Phule Pune University (SPPU)

- Final Year Project: [Alumni Management Portal](#) [\(live link\)](#) — Designed and developed a full-stack alumni management system that was officially adopted and deployed by the college. The portal is actively used by the institution for alumni tracking, networking, and engagement.
- College Core Committee: Served as Secretary, Refreshment & Food Department — Managed logistics, coordination, and execution of food and refreshment operations for all major college events and functions.

CERTIFICATIONS

- **Deep Learning Onramp** — MathWorks (2024)
- **Python Programming** — GUVI (2023)
- **Geodata Processing using Python** — ISRO (2023)
- **Global Navigation Satellite System (GNSS)** — ISRO (2023)